AT2

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage : 20%

QUESTIONS: **Total Marks: 50**

Annexure A

Scenario-Based

Code of Conduct:

I K.S. Dinesh (192324185) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Dinesh

Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher
A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.
B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System
A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.
B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network
Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.
B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security
An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.
B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices
A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.
B. Recommend a suitable cryptographic approach and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1) Encryption Algorithm for Banking system:

a) identify the most suitable Algorithm:

* The Most suitable algorithm among DES, 3-DES and Blowfish is 3-DES (Triple Data Encryption standard) because it Provide stronger security than DES.

* Banking Application handle sensitive information such as Customer details, account numbers and financial transactions. Therefore, encryption algorithm must Protect data from unauthorized access and attacks.

* 3-DES improves security Compared with DES, making it a better choice when Considering only these three algorithms. However, for modern banking system, AES is generally preferred because it Provides stronger security and better performance.

B) Justify your choice Based on Security Strength:

DES (Data Encryption Standard)

DES is a symmetric-key encryption algorithm that uses a 56 bit effective key. it encrypts

Security : DES is vulnerable to brute force attacks.

Efficiency : it is relatively fast on older system.

Application : it is unsuitable for protecting transactions.

3-DES (Triple Data Encryption standard) :

Security : It offers better protection than DES

Efficiency : Applying encryption three times.

Application : It was used in financial system.

Blowfish:

Blowfish is a symmetric-key encryption algorithm designed for fast encryption. it supports variable key length from 32 to 448 bits.

Comparison Table:

Feature	DES	3-DES	Blowfish
1. Key Length	56 bits	up to 168 bits	32-448 bits
2. Security	Weak	Stronger	Good
3. Speed	Fast	Slow	Fast
4. Size	64-bits	64 bits	64 bits

2) Public key cryptography for Email security:

A) How RSA is used for Encryption and Decryption

RSA is an asymmetric encryption algorithm that uses two keys: a Public key and Private key.

1. Key generation: Each user generates a Public key and a private key.

2. Encryption: The Sender encrypts the email message using the receipients Public key.

3. Transmission: The encrypted emails is sent internet.

4. Decryption: The recipient uses their private key to decrypt and read the email.

B) Role of Key Distribution and Management:

Key distribution and management are essential for maintaining email security.

1. Public Key distribution: Public keys can be shared through digital certificates.

2) Private key Protection : Private keys must be kept.

3) Digital Certificates : A trusted Certificate Authority (CA)

4) key renewal : keys should be renewed security.

5) Revocation : Compromised or invalid Certificates.

RSA Key Generation:

$$n = p \times q$$

Euler's totient : $\phi(n) = (p-1)(q-1)$

Private exponent $d \Rightarrow d \times e \equiv 1 \pmod{\phi(n)}$

RSA encryption formula :

$$c = m^e \bmod n$$

Decryption : $m = c^d \bmod n$

Example of Secure Email Communication:

Sender

↓

Public key

↓

Encrypted Email

↓

Private key

↓

Recipient